

Problem Statement

Disasternet AI - An Intelligent AI-Powered Disaster Response, Prediction, and Resource Management Platform

Natural and human-induced disasters such as floods, cyclones, earthquakes, wildfires, landslides, industrial accidents, and pandemics are becoming more frequent and severe due to climate change, rapid urbanization, and environmental degradation. Every year, millions of people are displaced, thousands lose their lives, and governments suffer billions of dollars in economic losses because emergency response systems are often fragmented, slow, and dependent on manual coordination.

The Objective

Build an AI-powered Disaster Response and Emergency Management Platform that delivers real-time disaster intelligence, emergency coordination, and AI-assisted decision support for citizens, rescue teams, healthcare organizations, NGOs, and government agencies.

- Real-Time Monitoring - Track disasters using weather APIs and GIS maps.
- AI Emergency Assistant - Provide instant safety guidance and evacuation support.
- Interactive Disaster Map - Display shelters, hospitals, rescue teams, and safe routes.
- Resource Management - Optimize allocation of food, medicine, rescue teams, and supplies.
- Predictive Risk Analysis - Identify high-risk areas using AI and weather data.
- Multi-Agent AI System:
 - Disaster Intelligence Agent
 - Risk Prediction Agent
 - Emergency Response Agent
 - Resource Management Agent
 - Citizen Assistance Agent
- Emergency Communication - Enable incident reporting and verified emergency alerts.
- Analytics Dashboard - Visualize disaster status, shelter occupancy, and rescue progress.